BrandGuard AI — Round 2 Project Report

Assessment date: 2026-11-05
Project location: `C:\CodeNexus`
Readiness: Functional local MVP; **not production-deployment verified**.

1. Introduction

BrandGuard AI is a brand-governance application for organisations that need to review marketing copy against
The application stores each organisation's guidelines, submissions, findings, correction decisions, and reviewer
workspace.

2. Problem statement and solution

Brand policies are often long documents that are hard to apply consistently during content creation. BrandGuar
proposes candidate rules for a human to review, retrieves only approved rules as analysis evidence, and guides
corrections and final approval. It does not publish content or claim to certify legal/factual correctness.

3. Round 2 feature-by-feature status

Module	Status	Verified implementation and remaining caveat
Authentication and organisation management	Implemented	Registration creates an organisation and admin; login uses JWT and Argon2; admin c management/review routes; tenant separation is covered by tests. Logout clears the c invitations, email verification, and password reset are not implemented.
Brand knowledge hub	Implemented	PDF, DOCX, and TXT guideline upload and extraction; version links; manually authore extraction; candidates remain pending until an admin/brand manager approves them. directive cues, not generative AI.
Content submission	Implemented	PDF, DOCX, and TXT extraction; original file is retained; extracted text preview, submi association are provided. OCR and structured marketing metadata (such as CTA/clai
RAG and retrieval	Implemented with provider limitation	ChromaDB indexes approved rules only, with organisation-specific collections and me partial-analysis notes. The only configured embedding implementation is Chroma's loc switching is not implemented.
Hybrid compliance analysis	Implemented with AI limitation	Deterministic prohibited-term/disclaimer checks run locally; optional Gemini structured submitted text and retrieved rule IDs. Gemini was not tested because no API key was verification is not implemented.
Compliance report	Implemented	Actual saved findings, evidence, analysis status, score, severity, confidence, correctio Score uses `severity-weighted-v1`: Critical 25, High 15, Medium 8, Low 3 points de review heuristic, not accuracy or legal certification.

		review heuristic, not accuracy or legal certification.
Correction suggestions	Implemented	Suggestions can be created, accepted, rejected, or edited. Suggested wording and de original submission text remains unchanged. Gemini-based suggestions are optional.
Human review	Implemented	Reviewer, decision, comment, timestamp, prior status, and resulting status are persist comment. Human approval remains final; publishing is not automated.
Dashboard and analytics	Implemented	Counts, severity distribution, compliance trends, and recent activity come from persist and finding-severity filters are available. Organisation is fixed to the authenticated tena filtering.
Security and data protection	Core controls implemented	Password hashing, JWT, role checks, organisation-scoped API access, upload validat environment configuration, CORS allowlist, and audit events are present. Rate limiting configured production gateway remain deployment responsibilities.

4. Architecture and technology

```
React + Vite + Tailwind + React Router
              │ Axios (/api proxy or configured API origin)
              ▼
FastAPI routers ── Pydantic schemas ── SQLAlchemy services
      │                    │
      │                    ├── PostgreSQL (deployment) / SQLite (local)
      │                    └── Alembic migrations
      ├── document extraction (PyMuPDF, python-docx, UTF-8 TXT)
      ├── ChromaDB per-organisation approved-rule collections
      └── optional Gemini structured analysis and suggestions
```

The frontend uses React 18, Vite, Tailwind CSS, React Router, Axios, Lucide, and Recharts. The backend uses SQLAlchemy, Alembic, PyMuPDF, python-docx, ChromaDB, and the Google GenAI SDK.

5. Data model and AI/RAG flow

Principal database records include `Organization`, `User`, `BrandGuideline`, `BrandRule`, `ContentSubmission`, `Violation`, `CorrectionSuggestion`, `ApprovalHistory`, `AuditLog`, and `BrandAsset`. Tenant-owned business ID. Guideline version records link to the prior version while keeping rules tied to their source document. The late review status transitions.

The analysis sequence is: extract text → identify draft source rules → human approval → index approved rules → run deterministic checks and optional Gemini → validate finding text/evidence IDs → persist report and score → human review and audit. Missing Gemini or retrieval support is reported as partial analysis, not a successful

6. Local installation and operation

Requirements: Python 3.11+, Node.js 20+, and optionally Docker/PostgreSQL for local database testing.

1. Copy `.env.example` to `.env` and configure a strong random `JWT_SECRET_KEY`. Set `DATABASE_URL` for P
   `GEMINI_API_KEY` is optional.
2. In `backend`, create/activate a virtual environment and run `python -m pip install -r requirements`
3. Apply migrations with `alembic upgrade head`, then start `uvicorn app.main:app --reload --port`
4. In `frontend`, run `npm install` and `npm run dev`.
5. Visit `http://localhost:5173`, register an admin, upload a guideline, extract/edit candidates, approve r
   and review.
6. API documentation is served at `http://localhost:8000/docs`.

The Vite development proxy targets `http://localhost:8000` by default. `VITE_API_PROXY_TARGET` changes t
use `VITE_API_BASE_URL`.

7. Deployment configuration and verification

Deployment scaffolding exists in `render.yaml`, `backend/Dockerfile`, `docker-compose.yml`, and `frontend/`
configured for PostgreSQL, persistent upload/Chroma storage, production environment validation, and a health
SPA and accepts the API base URL.

Not verified: no hosted PostgreSQL service, Render deployment, Vercel deployment, or configured Gemini key
deployment URL is claimed. Before public launch, configure production secrets/origins, verify database migratic
storage, set gateway rate limits, test provider calls, and run a deployed end-to-end workflow.

8. Verification evidence

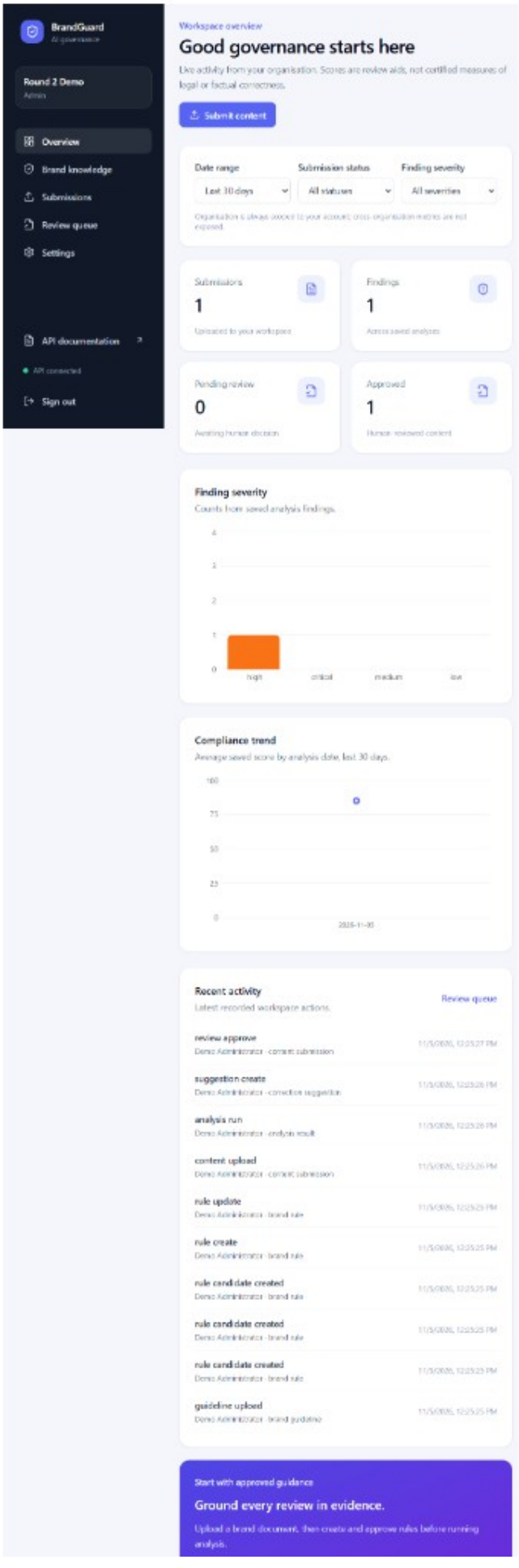
The local API was exercised end to end using a real demo organisation: TXT guideline upload, three source-de
explicit rule approval, content upload, evidence-linked partial analysis, correction suggestion, and human appro
The original submission was not altered.

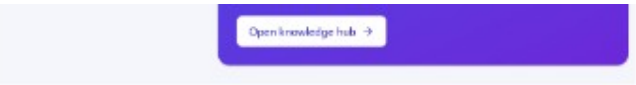
Automated checks run after the Round 2 changes:

- `pytest -q`: **19 passed**, one upstream Starlette/AnyIO deprecation warning.
- `npm run build`: succeeded.
- `npm audit`: **0 vulnerabilities**.
- `python -m compileall -q backend\app backend\alembic`: succeeded.
- `alembic upgrade head` and `alembic check`: succeeded; no schema drift.
- `.gitignore` checked: `.env`, uploads, Chroma data, database files, and `node_modules` are ignored.

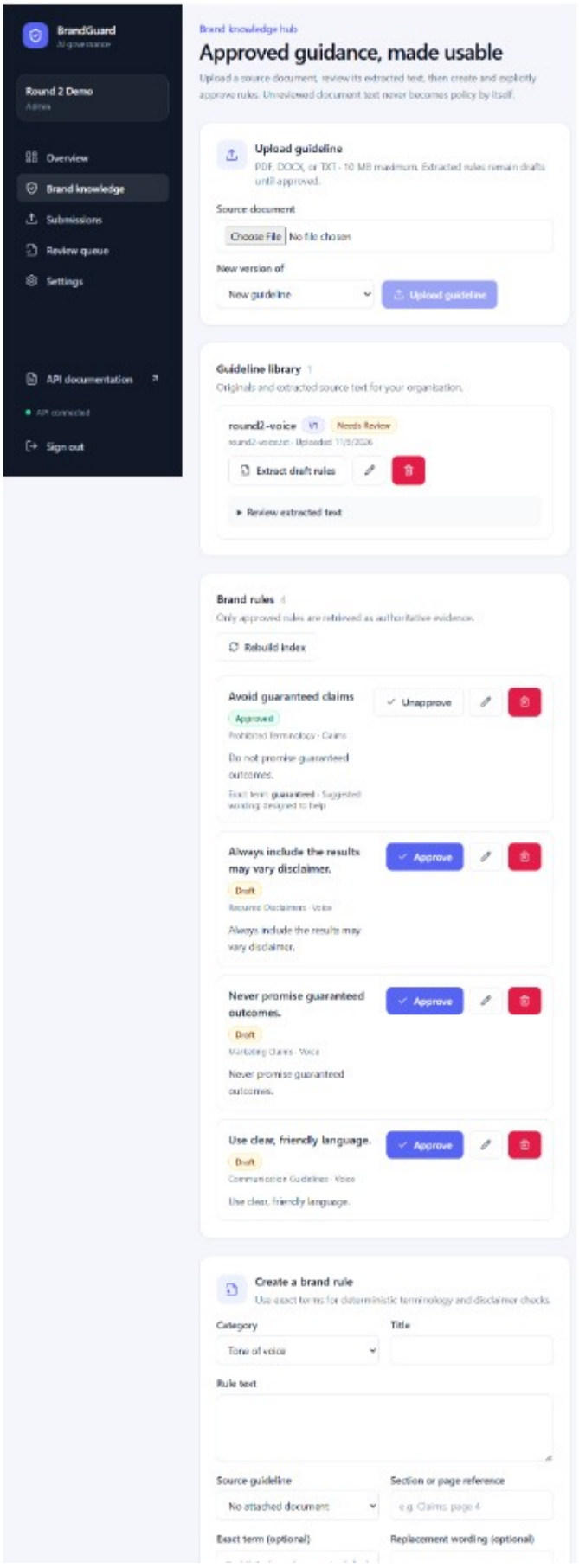
The screenshots below are actual captures of the locally running frontend backed by the local API. Demo data i
directories, not tracked source files.

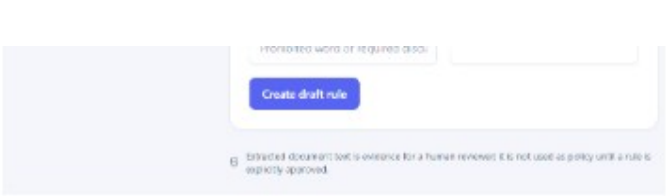
Dashboard


BrandGuard
Round 2 Demo
Admin
Overview
Brand knowledge
Submissions
Review queue
Settings
API documentation
API connected
Sign out
Workspace overview
Good governance starts here
Like activity from your organisation. Scores are review aids, not certified measures of legal or factual correctness.
Submit content
Date range
Last 30 days
Submission status
All statuses
Finding severity
All severities
Submissions
1
Findings
1
Pending review
0
Approved
1
Finding severity
Counts from saved analysis findings.
high
critical
medium
low
Compliance trend
Average saved score by analysis date, last 30 days.
Recent activity
Latest recorded workspace actions.
Review queue
review approve
suggestion create
analysis run
content upload
rule update
rule create
rule candidate created
rule candidate created
rule candidate created
guideline upload
Start with approved guidance
Ground every review in evidence.
Upload a brand document, then create and approve rules before running analysis.



Brand knowledge hub





Evidence-linked compliance report

